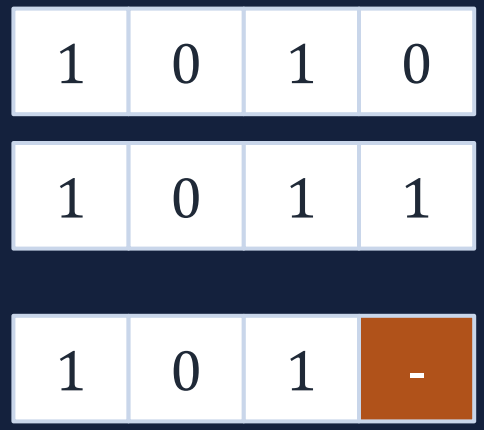
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 8

Tabular (Quine–McCluskey) Minimization

Implicants in dash notation · combining to prime implicants · the prime implicant chart · essential primes · cyclic charts and Petrick's method



$$m_{10} + m_{11} = AB'C$$

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Course supplement: tabular minimization

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Today’s roadmap

1

Why tabular?

An algorithm, not a picture

2

Prime implicants

Combine, then combine again

3

The PI chart

Essentials first

4

Cyclic charts

Petrick’s method

By the end of class you can…

- Write implicants in dash notation
- Find all prime implicants by combining tables
- Build a prime implicant chart and pick the essentials
- Finish a cover when essentials are not enough
- Solve a cyclic chart with Petrick’s method

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Agenda for today's session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap	—
0:05 – 0:15	Why a tabular method; dash notation	—
0:15 – 0:40	Finding the prime implicants	Example 1
0:40 – 0:50	Break	
0:50 – 1:10	Prime implicant chart; essentials	Example 1 cover
1:10 – 1:25	Full problem in pairs	Your turn
1:25 – 1:40	Cyclic charts and Petrick's method	Example 2
1:40 – 1:50	Concept check and wrap-up	Exam 1 preview

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Why a tabular method?

Limits of K-maps

Adjacency is hard to see past 4 variables and impractical past 6.

Choosing groups by eye can miss a better cover.

The tabular method

Same theorem as the map, $xy + xy' = x$, applied systematically.

Works for any number of variables and is easy to program.

1. **Step 1**: list minterms and don't-cares in binary, grouped by the number of 1s.
2. Combine terms in neighbouring groups that differ in one bit; repeat until nothing combines.
3. Terms never combined are the **prime implicants**.
4. **Step 2**: build the chart; take the **essential** primes.
5. Cover the remaining minterms with the fewest extra primes.

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Implicants in dash notation

Terms	Patterns	Result	Term
m_{10}, m_{11}	1010, 1011	101–	$AB'C$
m_8, m_{12}	1000, 1100	1–00	$AC'D'$
101–, 111–	differ in bit B	1–1–	AC
m_4, m_9	0100, 1001	—	3 bits differ
10–1, 1–10	dashes differ	—	cannot combine

The combining rule

Two terms combine when they differ in exactly one bit and their dashes are in the same places.

The differing bit becomes a dash: one fewer literal.

Only neighbouring groups (one more 1) can differ in one bit, so only those are compared.

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Example 1 — List by number of 1s

EXAMPLE

Minimize $F(A,B,C,D) = \Sigma m(4,8,10,11,12,15) + d(9,14)$ by the tabular method.

1s	Minterms	Pattern	
1	4	0100	✓
	8	1000	✓
2	(9)	1001	✓
	10	1010	✓
	12	1100	✓
3	11	1011	✓
	(14)	1110	✓
4	15	1111	✓

1. Include the don’t-cares 9 and 14: they may help combining.
2. Write each term in binary, A as the MSB.
3. Group by the number of 1s: 1, 2, 3 and 4.
4. Compare every term with every term in the next group.
5. Mark ✓ each term that combines; the rest are primes.

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Example 1 — Size-2 implicants

1s	Minterms	Pattern	
1	4,12	-100	PI
	8,9	100-	✓
	8,10	10-0	✓
	8,12	1-00	✓
2	9,11	10-1	✓
	10,11	101-	✓
	10,14	1-10	✓
	12,14	11-0	✓
3	11,15	1-11	✓
	14,15	111-	✓

1. Group 1–2: $4,12 \rightarrow -100$; $8,9 \rightarrow 100-$; $8,10 \rightarrow 10-0$; $8,12 \rightarrow 1-00$
2. Group 2–3: $9,11$; $10,11$; $10,14$; $12,14$
3. Group 3–4: $11,15 \rightarrow 1-11$; $14,15 \rightarrow 111-$
4. −100 combines with nothing: prime $BC'D'$
5. Every other size-2 term combines in the next column

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Example 1 — Size-4 and the primes

1s	Minterms	Pattern	
1	8,9,10,11	10--	PI
	8,10,12,14	1--0	PI
2	10,11,14,15	1-1-	PI

Prime implicant	Pattern	Covers
$BC'D'$	-100	4, 12
AB'	10--	8, 9, 10, 11
AD'	1--0	8, 10, 12, 14
AC	1-1-	10, 11, 14, 15

1. Combine size-2 terms whose dashes line up
2. 10–0 with 10–1 gives **10––**, and so on
3. The same size-4 term appears twice: list it once
4. No size-4 terms combine further
5. Four prime implicants in all

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Break

10 minutes — we resume at the 0:50 mark

Next: the prime implicant chart

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The prime implicant chart

Prime	Term	4	8	10	11	12	15
-100	BC′D′	×				×	
10--	AB′		×	×	×		
1--0	AD′		×	×		×	
1-1-	AC			×	×		×

Reading the chart

Columns: the minterms only (no don’t-cares).

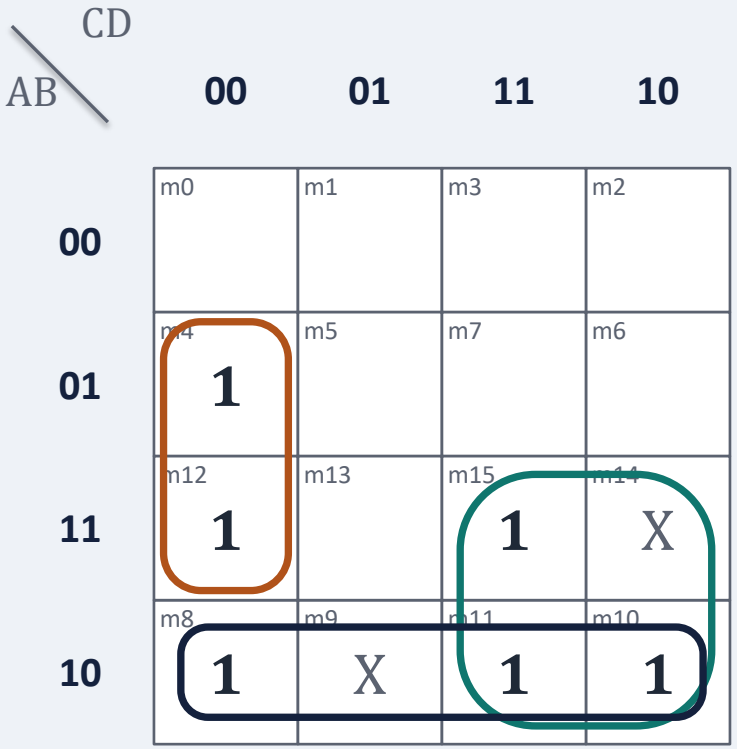
A column with a single × makes its row essential.

1. Column 4 has one ×: $BC'D'$ is essential
2. Column 15 has one ×: AC is essential
3. They cover 4, 12 and 10, 11, 15
4. Only minterm 8 is left
5. AB' or AD' covers it: either gives a minimum

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Example 1 — The minimum cover

Check on the map



1. Essentials: $BC'D'$ and AC
2. Add AB' (or AD') for minterm 8
3. $$F = BC'D' + AC + AB'$$
4. or $F = BC'D' + AC + AD'$: same cost
5. 7 literals, 3 terms: matches the map

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Your turn — A full problem

YOUR TURN ≈ 15 min

Minimize $F(A, B, C, D) = \Sigma m(0,1,2,5,6,7,8,9,10,14)$ by the tabular method.

(a) Build the combining columns and list the prime implicants.

(b) Draw the prime implicant chart and mark the essentials.

(c) Write the minimum sum of products.

Six prime implicants exist. Check your answer on a K-map.

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Solution — Combining columns

SOLUTION

1s	Minterms	Pattern	
0	0	0000	✓
1	1	0001	✓
	2	0010	✓
	8	1000	✓
2	5	0101	✓
	6	0110	✓
	9	1001	✓
	10	1010	✓
3	7	0111	✓
	14	1110	✓

1s	Minterms	Pattern	
0	0,1	000-	✓
	0,2	00-0	✓
	0,8	-000	✓
1	1,5	0-01	PI
	1,9	-001	✓
	2,6	0-10	✓
	2,10	-010	✓
	8,9	100-	✓
	8,10	10-0	✓
2	5,7	01-1	PI
	6,7	011-	PI
	6,14	-110	✓
	10,14	1-10	✓

1s	Minterms	Pattern	
0	0,1,8,9	-00-	PI
	0,2,8,10	-0-0	PI
1	2,6,10,14	--10	PI

1. Column 1: the ten minterms, grouped by their number of 1s.
2. Column 2: join terms in neighbouring groups that differ in one bit; ✓ both.
3. Column 3: join column-2 terms whose dashes line up.
4. Rows marked PI never combined: they are the prime implicants.

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Solution — The prime implicants

SOLUTION

Prime implicant	Pattern	Covers	Column
$A'C'D$	0-01	1, 5	size 2
$A'BD$	01-1	5, 7	size 2
$A'BC$	011-	6, 7	size 2
$B'C'$	-00-	0, 1, 8, 9	size 4
$B'D'$	-0-0	0, 2, 8, 10	size 4
CD'	--10	2, 6, 10, 14	size 4

1. Column sizes: 10, 13, 3
2. Three size-4 primes: $B'C'$, $B'D'$, CD'
3. Three size-2 primes remain unused
4. $A'C'D$, $A'BD$, $A'BC$
5. Six primes in all

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Solution — Chart and cover

SOLUTION

Prime	Term	0	1	2	5	6	7	8	9	10	14
0-01	A′C′D		×		×						
01-1	A′BD				×		×				
011-	A′BC					×	×				
-00-	B′C′	×	×					×	×		
-0-0	B′D′	×		×				×		×	
--10	CD′			×		×				×	×

1. Columns 9 and 14 each have a single ×.
2. So $B'C'$ and CD' are **essential**.
3. They cover 0, 1, 2, 6, 8, 9, 10, 14.
4. Left to cover: minterms 5 and 7.
5. $A'BD$ covers both ($A'C'D$ and $A'BC$ cover one each).
6. $F = B'C' + CD' + A'BD$ (7 literals)

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Cyclic charts and Petrick's method

Cyclic chart

Every column has at least two ×: there are no essential primes to start from.

Petrick's method

Name the primes $P_1, P_2, \ldots$. For each minterm write the OR of the primes that cover it.

AND all those sums, multiply out, and absorb: each product is a complete cover.

Pick the product with the fewest (then cheapest) primes.

Useful reductions

$(P_1 + P_2)(P_1 + P_3) = P_1 + P_2P_3$

Row dominance: drop a prime whose × are a subset of another's at no lower cost.

Column dominance: drop a minterm column whose × include another column's.

CAD tools use these steps with branching for large problems.

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Example 2 — A cyclic chart

EXAMPLE

Minimize $F(A,B,C) = \Sigma m(0,1,2,5,6,7)$.

	Prime	Term	0	1	2	5	6	7
P1	00-	A′B′	×	×				
P2	0-0	A′C′	×		×			
P3	-01	B′C		×		×		
P4	-10	BC′			×		×	
P5	1-1	AC				×		×
P6	11-	AB					×	×

1. Six primes, each covering 2 minterms
2. Every column has two ×: no essentials
3. Each column needs one of its two primes:
4. $P = (P_1 + P_2)(P_1 + P_3)(P_2 + P_4)(P_3 + P_5)(P_4 + P_6)(P_5 + P_6)$
5. Multiply out and absorb (next slide)

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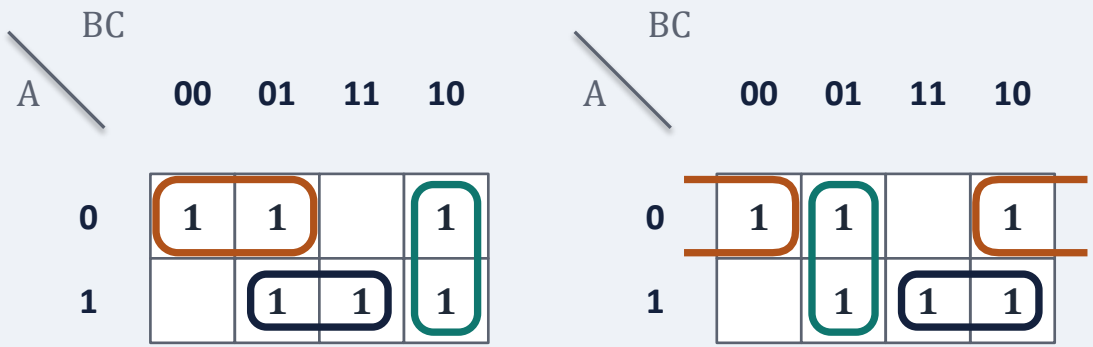
Example 2 — Petrick's method

1. One sum per minterm: 0 $(P_1 + P_2)$, 1 $(P_1 + P_3)$, 2 $(P_2 + P_4)$, 5 $(P_3 + P_5)$, 6 $(P_4 + P_6)$, 7 $(P_5 + P_6)$
2. Pair sums that share a prime: $(P_1 + P_2)(P_1 + P_3) = P_1 + P_2P_3$
3. $(P_2 + P_4)(P_4 + P_6) = P_4 + P_2P_6;\ (P_3 + P_5)(P_5 + P_6) = P_5 + P_3P_6$
4. Multiply the first two: $P_1P_4 + P_1P_2P_6 + P_2P_3P_4 + P_2P_3P_6$
5. Multiply by $(P_5 + P_3P_6)$ and use $XX = X$:
   $P_1P_4P_5 + P_1P_3P_4P_6 + P_1P_2P_5P_6 + P_1P_2P_3P_6 + P_2P_3P_4P_5 + P_2P_3P_4P_6 + P_2P_3P_5P_6 + P_2P_3P_6$
6. Absorb: $P_2P_3P_6$ absorbs every product that contains it
7. Left: $P_1P_4P_5 + P_2P_3P_6 + P_1P_3P_4P_6 + P_1P_2P_5P_6 + P_2P_3P_4P_5$
8. Fewest primes: $P_1P_4P_5$ and $P_2P_3P_6$, 3 each

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Example 2 — The two covers

Two minimum covers



$A'B' + BC' + AC$

$A'C' + B'C + AB$

1. $P_1P_4P_5$: $P_1 = A'B'$, $P_4 = BC'$, $P_5 = AC$
2. $F = A'B' + BC' + AC$
3. $P_2P_3P_6$: $P_2 = A'C'$, $P_3 = B'C$, $P_6 = AB$
4. $F = A'C' + B'C + AB$
5. Each map: every 1 is covered by the three groups ✓
6. Both: 3 terms, 6 literals: either is a minimum

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Tabular method or K-map?

	K-map	Tabular method
Variables	up to 4 (5–6 with effort)	any number
Method	visual pattern finding	systematic algorithm
Don’t-cares	used when grouping	used in step 1 only
Prime implicants	found by eye	all found, guaranteed
Choosing a cover	by eye	chart, then Petrick
Best for	hand work, insight	computers, large problems

The number of prime implicants can grow exponentially, so real CAD tools add heuristics.

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Concept check

1. Why are only neighbouring groups (by count of 1s) compared?

2. Why are don’t-cares used in step 1 but not in the chart?

3. What makes a prime implicant essential?

4. What does it mean when every chart column has two or more ×?

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Key takeaways

Combine

$$xy + xy' = x$$

Dash notation

$$101- = AB'C$$

Prime implicant

never combined

Essential

only × in a column

Cyclic chart

Petrick’s method

Don’t-cares

step 1 only

Same theorem as the K-map, applied by algorithm.

Homeworks #4 and #5 due at the start of Lecture 9

This afternoon’s lab: Lab C — Gate-level Verilog, propagation delay, testbenches

Next: review for Exam 1

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